
GLONET: MERCATOR’S END-TO-END NEURAL FORECASTING SYSTEM

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ABSTRACT

Accurate ocean forecasting is crucial in different areas ranging from science to decision making. Recent advancements in data-driven models have shown significant promise, particularly in weather forecasting community, but yet no data-driven approaches have matched the accuracy and the scalability of traditional global ocean forecasting systems that rely on physics-driven numerical models and can be very computationally expensive, depending on their spatial resolution or complexity. Here, we introduce GLONET, a global ocean neural network-based forecasting system, developed by Mercator Ocean International. GLONET is trained on the global Mercator Ocean physical reanalysis GLORYS12 to integrate physics-based principles through neural operators and networks, which dynamically capture local-global interactions within a unified, scalable framework, ensuring high small-scale accuracy and efficient dynamics. GLONET’s performance is assessed and benchmarked against two other forecasting systems: the global Mercator Ocean analysis and forecasting 1/12° high-resolution physical system GLO12 and a recent neural-based system also trained from GLORYS12. A series of comprehensive validation metrics is proposed, specifically tailored for neural network-based ocean forecasting systems, which extend beyond traditional point-wise error assessments that can introduce bias towards neural networks optimized primarily to minimize such metrics. The preliminary evaluation of GLONET shows promising results, for temperature, sea surface height, salinity and ocean currents. GLONET’s experimental daily forecast are accessible through the European Digital Twin Ocean platform EDITO.

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1 Introduction

The accurate prediction of oceanic states is crucial for numerous marine operations, encompassing navigation [James, 1957], fisheries management [Pikitch et al., 2004], disaster response [Teal and Howarth, 1984, Breivik et al., 2013, Zohdi and Abbaspour, 2019], and climate research. Ocean forecasting has primarily depended on physics-based numerical ocean models [Rosati and Miyakoda, 1988, Madec et al., 1997, Chassignet et al., 2007] co-developed and operated by various operational forecasting centers worldwide. These numerical models employ intricate systems of partial differential equations to simulate ocean dynamics, sometimes with the constraint of not departing too far from both contemporary and historical oceanographic observations. Operational Global Ocean Forecasting Systems (GOFs) merge these numerical frameworks with real-time observational database via advanced data assimilation techniques [Bannister, 2017, Brasseur et al., 2005, Cummings and Smedstad, 2013, Lellouche et al., 2018, Copernicus, 2023] to deliver precise and timely forecasts. However, these models face significant challenges: they are computationally expensive, slow to improve, and require substantial resources for forecasting the state of the ocean, limiting their ability to provide forecasts rapidly, especially in situations that require high-frequency updates or providing ensemble forecasts and their associated uncertainty. This sluggishness also hampers their use in research, where exploring a wide range of scenarios quickly and efficiently can be essential for advancing ocean science. Moreover, even with the influx of more observational data, it is often challenging to improve these models due to their reliance on fixed physical assumptions and the need for extensive parameter tuning. The need for faster, more flexible approaches becomes then evident as traditional models struggle to adapt to evolving requirements and increasingly diverse data sources [Liu et al., 2017].

In recent years, the weather forecasting community has witnessed a significant surge in the development and deployment of neural network (NN)-based forecasting systems. These data-driven models leverage advancements in machine learning, particularly deep learning, to enhance forecast accuracy and computational efficiency. Weather forecasting models such as FourCastNet [Pathak et al., 2022], GraphCast [Lam et al., 2022], Pangu-Weather [Bi et al., 2023] and recently AIFS [Lang et al., 2024] have demonstrated remarkable improvements in prediction performance and speed, often outperforming traditional numerical weather prediction (NWP) methods in various metrics [Rasp et al., 2024]. The rapid adoption of NN-based systems in meteorology underscores the potential of artificial intelligence (AI) to revolutionize environmental forecasting.

On the other hand, the oceanographic community faces distinct challenges in developing similar AI-driven forecasting systems. The ocean environment is inherently more complex due to factors like continental boundaries, islands, complex boundary conditions, varying water mass properties, and intricate ocean-atmosphere interactions. These complexities make it significantly more difficult to create data-driven models that can accurately capture the spatial-temporal dynamics of the global ocean. Additionally, the state of AI-based ocean forecasting remains nascent, with limited models achieving the high-resolution forecasts necessary for operational use, typically at $1/4^\circ$ or finer, further exacerbates the computational demands, making the training and deployment of deep learning models particularly challenging.

Addressing these challenges, Mercator Ocean International (MOI) has developed a novel neural network-based forecasting system, GLONET, which represents a significant advancement in the field of ocean forecasting. GLONET is trained on the $1/12^\circ$ daily mean outputs of the global Mercator Ocean physical reanalysis GLORYS12 [Lellouche et al., 2021] and is designed to operate at a horizontal resolution of $1/4^\circ$ while inheriting fine-scale dynamics of GLORYS12, enabling it to resolve mesoscale oceanic features with unprecedented accuracy. Leveraging a hierarchical transformer-based backbone, GLONET incorporates physics-based principles through neural operators and networks to dynamically capture local-global interactions within a unified, scalable framework, mitigating the complexities introduced by continental boundaries, and ensuring high small-scale accuracy and efficient dynamics over short forecast intervals.

To ensure the operational viability of GLONET, MOI has established robust pre-operational pipelines that facilitate its integration into existing forecasting workflows. Additionally, GLONET undergoes rigorous validation using the Class-4 framework [Ryan et al., 2015, Hernandez et al., 2009, Divakaran et al., 2015, Lellouche et al., 2013], a stringent evaluation protocol that assesses forecast accuracy against leading operational GOFs. In addition to traditional metrics, GLONET also undergoes a unique NN-specific evaluation criterion, proposed in this paper, to meet the high standards required for operational deployment. In addition, using the aforementioned validation frameworks, GLONET is benchmarked against the current global Mercator Ocean analysis and forecasting $1/12^\circ$ high-resolution physical system GLO12 [Lellouche et al., 2023], and Xihe, one of the first neural global ocean forecasting system [Wang et al., 2024] trained from the same ocean reanalysis GLORYS12.

The paper is organized as follows. Section 2 gives an overview of the proposed methodology highlighting its different modules designed to enhance forecast accuracy. Section 3 details the experimental datasets employed in the $1/4^\circ$ configuration GLONET along with training. Section 4 presents experimental results, benchmarking GLONET’s performance against GLO12 and Xihe, using various validation metrics. Finally, Section 5 concludes the paper with a discussion of the implications of GLONET’s performance and potential avenues for future research.

2 Model

A diverse range of deep learning architectures such as vision transformers, graph neural networks, and neural operators has found application in data-driven weather forecasting and, more recently, in ocean forecasting [Wang et al., 2024], where vision and Swin transformers [Khan et al., 2022, Liu et al., 2021] were combined to develop a global ocean forecasting system. Building on these advances, we introduce GLONET, a multi-scale forecasting system. The ocean is fundamentally a turbulent system, marked by the interplay of processes across a wide range of spatial and temporal scales. To address this complexity, GLONET leverages Fourier neural operators [Li et al., 2020] to model broad, basin-wide patterns like gyres, currents, tele-connection and global circulation, capturing large-scale dependencies effectively representing the large-scale energy transfer that drives the background state of the ocean. At the small-scale dynamics, GLONET employs convolutional neural networks (CNNs) [O’Shea, 2015] to enhance predictions of submesoscale features, which influence localized mixing and fine-scale dependencies. Additionally, GLONET employs an encoder-decoder [Schmidhuber, 2015] to fuse these multi-scale circulations into a unified latent space, capturing essential detail across different scales that play a key role in redistributing heat, salt, and energy.

Let $D \subset \mathbb{R}^d$ define a spatial domain, with input function space $\mathcal{X} = \mathcal{X}(D; \mathbb{R}^{d_x})$ and the output space $\mathcal{Y} = \mathcal{Y}(D; \mathbb{R}^{d_y})$, respectively for the initial condition and forecast states, with $\mathbb{R}^{d_y} \in \mathbb{R}^{d_x}$.

GLONET aims to approximate the nonlinear mapping $\mathcal{G}^\dagger : \mathcal{X} \rightarrow \mathcal{Y}$ that forecasts the ocean state Y_{t+dt} given initial conditions X_t . Let $\{(\mathbf{X}_j, \mathbf{Y}_j)\}_{j=1}^N$ be our observed input-forecast pairs, where $\mathbf{X}_j = X_t^j$ represents the initial conditions and $\mathbf{Y}_j = Y_{t+dt}^j$ the target state. The aim is then to approximate $\mathcal{G}^\dagger$ using a parameterized operator $\mathcal{G}_\theta : \mathcal{X} \rightarrow \mathcal{Y}$, where $\theta \in \Theta$ denotes the parameter set, by minimizing a forecast error $\mathcal{L} : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$:

$$\min_{\theta \in \Theta} \mathbb{E}_{\mathbf{X} \sim \mu} [\mathcal{L}(\mathcal{G}_\theta(\mathbf{X}), \mathcal{G}^\dagger(\mathbf{X}))] \quad (1)$$

where μ is a probability measure over the initial conditions function space $\mathcal{X}$.

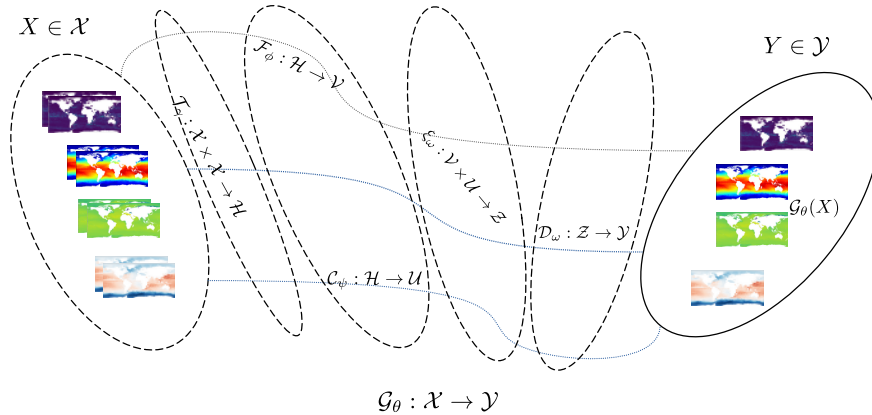


Figure 1: Overview of GLONET’s architecture containing different modules, particularly time-block designed to learn feature maps encapsulating initial conditions along with forcings. A spatial module architected to learn multi-scale dynamics, and finally an encoder-decoder to fuse multi-scale circulations into a unified latent space.

The operator $\mathcal{G}_\theta$ is architected to capture the multi-scale dynamics of ocean forecasting through a series of interconnected components as shown in Figure 1. It starts by processing two consecutive ocean states, $\mathbf{X}_{t-1}$

and $\mathbf{X}_t$, to extract meaningful feature maps that encapsulate both the initial condition along with the forcings influencing the ocean dynamics as follow:

$$\mathbf{H}_t = \mathcal{T}_\gamma(\mathbf{X}_{t-1}, \mathbf{X}_t), \quad (2)$$

where:

- $\mathcal{T}_\gamma : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{H}$ is the temporal encoding operator parameterized by γ .
- $\mathcal{H}$ represents the feature space that includes both the current ocean state and the learned forcings.

This block effectively integrates temporal information and forcings, providing a rich representation for subsequent spatial processing. To capture the diverse spatial scales inherent in the ocean, $\mathcal{G}_\theta$ employs processing through two distinct modules $\mathcal{F}_\phi$ and $\mathcal{C}_\psi$. The first is designed to model large-scale, basin-wide ocean patterns by capturing global dependencies and long-range interactions, providing a foundational representation of the ocean state. It operates in the Fourier domain, leveraging the Fourier transform $\mathcal{F}$ and its inverse $\mathcal{F}^{-1}$ to efficiently capture spectral information as:

$$\mathbf{v}_{t+1} = \mathcal{F}_\phi(\mathbf{H}_t) = \mathcal{F}^{-1}(\sigma(W \cdot \mathcal{F}(\mathbf{H}_t))), \quad (3)$$

with $W \in \mathbb{C}^{k_{\max} \times d_H \times d_H}$ are learned weights in the Fourier space, σ is a non-linear activation function applied element-wise, and $k_{\max}$ defines the maximum frequency mode considered. Simultaneously, $\mathcal{C}_\psi$ focuses on capturing small-scale, localized features and interactions. It operates directly in the spatial domain, applying convolutional filters to extract fine-grained patterns and defined as:

$$\mathbf{u}_{t+1} = \mathcal{C}_\psi(\mathbf{Y}_t), \quad (4)$$

where $\mathcal{C}_\psi : \mathcal{F} \rightarrow \mathcal{U}$ is a convolutional neural networks parameterized by ψ , and $\mathcal{U}$ the feature space capturing localized dependencies. This part enhances the model's ability to predict detailed, high-resolution oceanic phenomena that are critical for accurate localized forecasts. To integrate the large-scale and small-scale feature representations, $\mathcal{G}_\theta$ employs an encoder-decoder that fuses multi-scale circulations from $\mathbf{v}_{t+1}$ and $\mathbf{u}_{t+1}$ into a unified latent representation:

$$\mathbf{z}_{t+1} = \mathcal{E}_\omega(\mathbf{v}_{t+1}, \mathbf{u}_{t+1}), \quad (5)$$

where:

- $\mathcal{E}_\omega : V \times U \rightarrow Z$ is the encoder operator parameterized by ω .
- Z represents the latent feature space that encapsulates both large-scale and small-scale dynamics.

On the other hand, $\mathcal{D}_\omega$ maps the latent representation $\mathbf{z}_{t+1}$ back to the original spatial domain, generating the forecasted ocean state $\mathbf{X}_{t+dt}$:

$$\mathbf{X}_{t+dt} = \mathcal{D}_\omega(\mathbf{z}_{t+1}), \quad (6)$$

with $\mathcal{D}_\omega : Z \rightarrow \mathcal{Y}$ is a decoder operator parameterized by ω .

Combining all components, $\mathcal{G}_\theta$ can be expressed as the composition of all the aforementioned operators:

$$\mathcal{G}_\theta(\mathbf{X}_{t-1}, \mathbf{X}_t) = \mathcal{D}_\omega(\mathcal{E}_\omega(\mathcal{F}_\phi(\mathcal{T}_\gamma(\mathbf{X}_{t-1}, \mathbf{X}_t)), \mathcal{C}_\psi(\mathcal{T}_\gamma(\mathbf{X}_{t-1}, \mathbf{X}_t)))), \quad (7)$$

where $\theta = \{\gamma, \phi, \psi, \omega\}$ encapsulates all model parameters across the different modules. This composite mapping effectively captures the multi-scale dynamics of the ocean by integrating temporal dependencies, large-scale circulations, and localized features into a unified forecasting framework.

To efficiently handle the high-resolution requirements of GLONET, a multi-GPU pipeline that leverages pipeline parallelism is adopted, in which distinct segments of the model are distributed across different GPUs, passing activations between devices as the model progresses through the pipeline. This approach ensures that the computational load is shared and enables scaling to larger model sizes while accommodating high-resolution data. Additionally, activation checkpointing is employed throughout the forward pass to further reduce memory usage. Instead of keeping intermediate activations, which can quickly consume GPU memory, intermediate results are selectively recomputed during the backward pass. This strategy significantly minimizes

the memory footprint and reduces the peak memory requirement, allowing each GPU to handle larger parts of the model without exceeding memory limitations. Furthermore, to facilitate the above, GPU operations are synchronized using efficient communication primitives, enabling a smooth transfer of activations between GPUs with minimal overhead. By distributing both the model parameters and activations across multiple devices, GLONET is trained on fine-resolution datasets without compromising computational efficiency.

3 Training

GLONET is trained on the daily mean outputs of the GLORYS12 reanalysis and is configured to produce daily forecasts by using the ocean state inputs as shown in Figure 2. To optimize computational efficiency, GLONET operates on a coarser $[1/4]^\circ$ grid, by interpolating GLORYS12 products to this resolution. Despite this reduction in grid resolution, the model is designed to retain and represent the small-scale ocean dynamics intrinsic to the original $[1/12]^\circ$ products. This approach allows GLONET to balance the computational tractability of daily forecasts with the need to accurately capture mesoscale and finer processes, thus providing a nuanced representation of ocean dynamics without the high computational cost associated with finer grids.

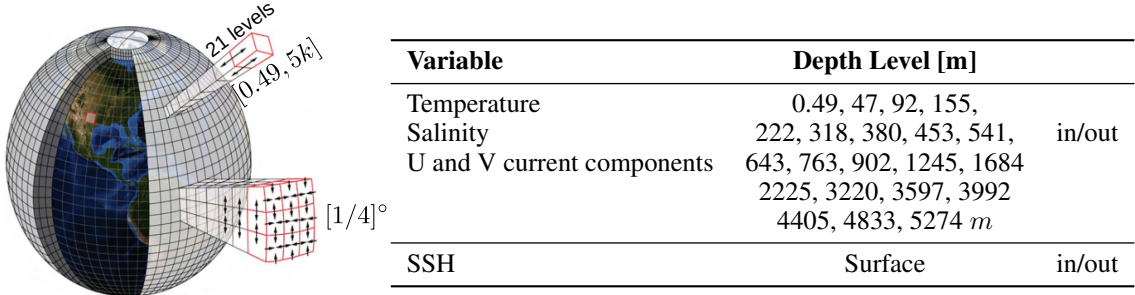


Figure 2: GLONET’s configuration, characterized by $[1/4]^\circ$ of spatial resolution and 21 vertical levels ranging from 0.4 to 5274 m of depth. U and V correspond to the zonal and meridional velocity respectively. SSH corresponds to the sea surface height.

GLONET forecasts the three-dimensional ocean state at t_{+1day} by leveraging the ocean states at t and t_{-1day} , thanks to its temporal module $\mathcal{T}_\gamma$, which integrates informative feature maps that encapsulate both the initial conditions and the external forcings modulating ocean dynamics. Glonet $\mathcal{G}_\theta(\mathbf{X}_{t-1}, \mathbf{X}_t) = \mathbf{X}_{t+1}$ is designed to provide 1 lead-day forecast. Therefore, longer lead-time forecasts are produced through an auto-regressive approach, where the GLONET iteratively inputs its own predictions. To enhance forecast accuracy, the same process is also considered during the training phase, where an initial pre-training phase focuses on generating 1-day-ahead forecasts, followed by a subsequent phase in which the GLONET extends forecasts up to 4 days. During backpropagation, gradients propagate across the entire 4-days forecast sequence, ensuring that temporal dependencies are captured and optimized throughout the training process.

The training of GLONET involves optimizing the model parameters θ to minimize the expected forecast error over the distribution μ of initial conditions, derived from GLORYS12 reanalysis. The optimization problem is formulated as:

$$\min_{\theta \in \Theta} \mathbb{E}_{\mathbf{X} \sim \mu} [\mathcal{L}(\mathcal{G}_\theta(\mathbf{X}), \mathcal{G}^\dagger(\mathbf{X}))], \quad (8)$$

where $\mathcal{L}$ is a suitable loss function (e.g., mean squared error) that quantifies the discrepancy between the predicted state $\mathcal{G}_\theta(\mathbf{X})$ and the true state $\mathcal{G}^\dagger(\mathbf{X})$. In practice, this expectation is approximated using an empirical average over a finite training dataset, sampled according to μ and based on GLORYS12 daily mean states:

$$\min_{\theta \in \Theta} \frac{1}{N} \sum_{j=1}^N \mathcal{L}(\mathcal{G}_\theta(\mathbf{X}_j), \mathbf{Y}_j). \quad (9)$$

This empirical risk minimization enables GLONET to learn a robust and generalizable forecasting model, accurately predicting the ocean state under varying initial conditions derived from high-resolution reanalysis data. The GLORYS12 outputs provide an extensive and consistent dataset, facilitating the model’s ability to capture the essential small and large-scale ocean dynamics needed for reliable forecasts.

Training was performed on $1/4^\circ$ interpolated GLORYS12 daily mean outputs for the period [1993, 2019], using Adam’s optimizer [Kingma, 2014], and a learning rate decreasing from e^{-4} to e^{-5} . Once the model is trained and the desired performances were reached, a second round of fine-tuning is performed to optimize the forecasts accuracy up to 4 days. The overall training took around 3 weeks while using a total of 32 40G-A100 GPUs. GLONET is capable of outputting 10 days forecast in less than 10 seconds.

4 Results

In this section, GLONET’s forecasting capabilities are empirically assessed using a comprehensive set of validation metrics, designed to capture different dimensions of forecast quality. The evaluation begins with the IV-TT Class 4 Framework, a robust benchmark that leverages observational data to evaluate forecasting accuracy against reference datasets. This is followed by point-wise accuracy assessments, process-oriented metrics, and diagnostic analyses, offering a multifaceted understanding of the model’s performance. Point-wise evaluations use GLORYS12 as the reference dataset, assessing the models’ ability to generate forecasts that align closely with the data on which they were trained. This analysis provides a detailed view of the models’ accuracy in capturing ocean states across various spatial and temporal scales, ensuring consistency and comparability with the training reference. Extending beyond these localized evaluations, process-oriented metrics are incorporated to evaluate the model’s ability to derive essential ocean quantities and maintain coherence across interconnected forecasted variables. Additionally, diagnostic variables, such as vorticity and mesoscale eddies are analysed to assess the model’s skill in preserving dynamic consistency, energy cascading and fine-scale structures. This holistic evaluation ensures that GLONET’s forecasts not only minimize numerical errors but also adhere to the fundamental dynamics governing ocean systems.

4.1 IV-TT Class 4 Framework Evaluation

In this part, GLONET, XIHE and GLO12 are evaluated using the CLASS4 dataset, developed and maintained by the "OceanPredict Task Team for Intercomparison and Validation" (IV-TT team). The CLASS4 framework provides a benchmark for model validation by operating within the observation space, enabling a direct comparison between observed and modeled values across both spatial and temporal dimensions. For each observation, the corresponding model counterpart is extracted at the same spatial and temporal location across various forecast lead times, ranging from the best analysis (day 0) to ten-day forecasts. The CLASS4 dataset includes observations of temperature and salinity from Argo profiles, sea surface temperature (SST) from surface drifting buoys, and sea level anomaly (SLA) from along-track satellite measurements. This framework serves as a robust tool for intercomparison, facilitating a comprehensive assessment of the forecasting models’ performance.

Figure 3 illustrates the dispersion and evolution of the Root Mean Square Difference (RMSD) as a function of forecast lead time for the following variables: temperature and salinity (5–100 m layer), SLA, and SST. XIHE performs comparably to GLO12 and better than GLONET in forecasting temperature in the 5–100 m layer. The three models exhibit similar performance in salinity, though GLONET shows greater dispersion for the 3- and 5-day forecasts. GLONET surpasses XIHE in SLA, consistently outperforming GLO12 for forecasts ranging from 5 to 9 days. For SST, GLO12 demonstrates superior performance across all forecast horizons, except for the one-day forecast, where the three models yield identical results.

To further explore regional performance differences, Figure 4 presents the spatial distribution of normalized Mean Absolute Error (MAE) differences. Here, the MAE is first computed for GLO12 as a reference, followed by the MAE calculations for GLONET and XIHE. Each map shows the difference between the reference and the model, normalized by the reference MAE. Positive values (red) indicate areas where the model outperforms GLO12, while negative values (blue) represent regions of degradation relative to the reference.

The neural network-based models, GLONET and XIHE, generally perform well in dynamically active regions such as the Kuroshio, Gulf Stream, Antarctic Circumpolar Current (ACC), and parts of the North Atlantic and North Pacific. However, both models show reduced performance in the tropical band and within subtropical gyres. GLONET outperforms XIHE in the South Pacific, potentially due to better generalization in regions with sparse observations or lower variability in training data. Conversely, XIHE demonstrates slightly better performance in the North Atlantic, likely benefiting from denser observational data in this well-monitored region.

When compared to GLO12, the neural network models reveal nuanced strengths and weaknesses. GLONET and XIHE may struggle in regions dominated by strong seasonal or sub-surface dynamics, as these are more

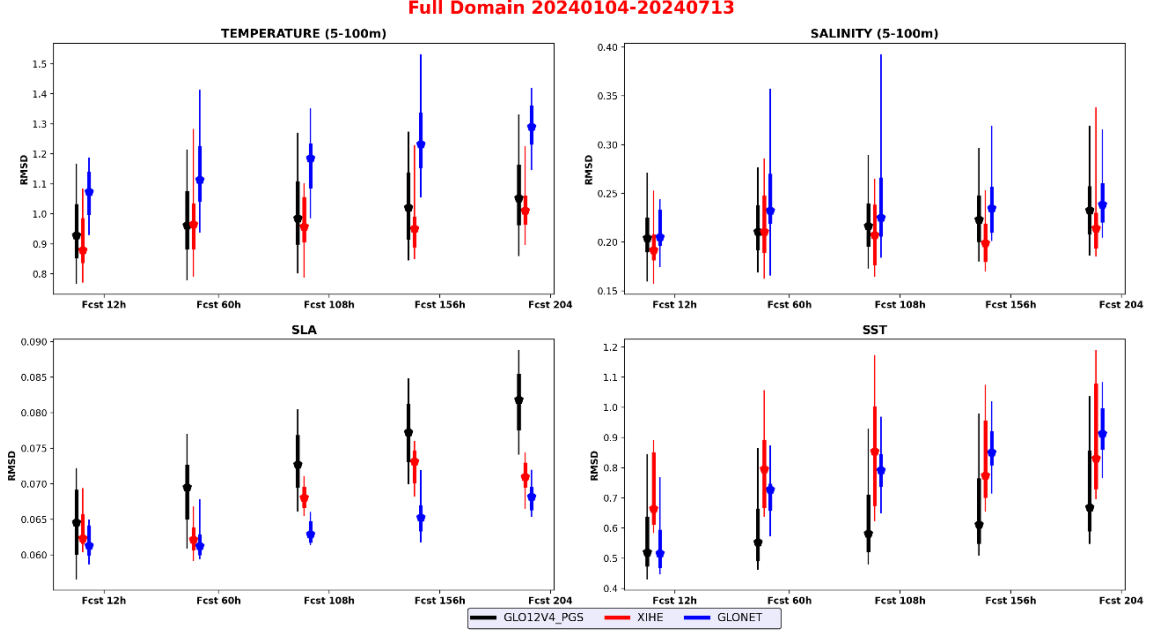


Figure 3: Dispersion and evolution of RMSD as a function of forecast lead time for temperature (top left) and salinity (top right) in the 5–100 m layer, SLA (bottom left), and SST (bottom right) from drifting buoys. The thick lines represent the 75% distribution, while the thin lines correspond to the 95% distribution.

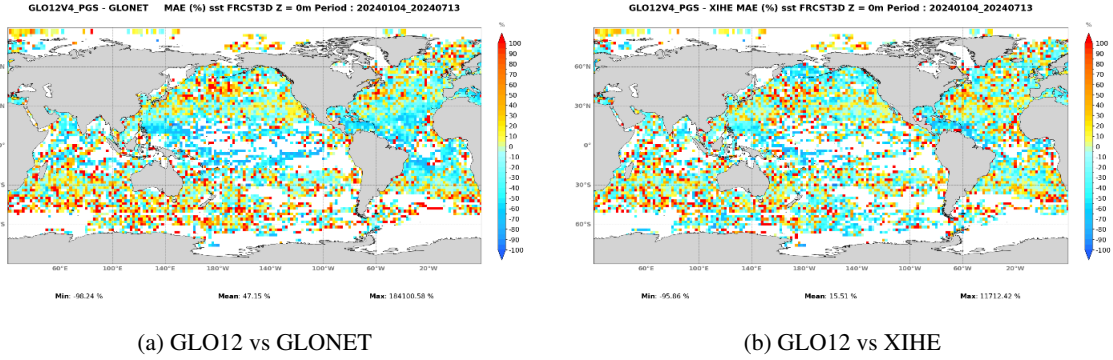


Figure 4: Normalized MAE difference between GLO12 and GLONET (left) and between GLO12 and XIHE (right). Positive values (red) denote improvement relative to GLO12, while negative values (blue) indicate degradation.

explicitly represented in physics-based models like GLO12. However, in regions characterized by mesoscale variability, such as the Kuroshio and Gulf Stream, the data-driven approaches can leverage their flexibility to capture localized patterns, occasionally outperforming GLO12. These results highlight the complementary strengths of neural network-based and physics-based approaches in forecasting regional ocean dynamics.

4.2 Point-wise evaluation

In this part, the point-wise forecasting accuracy of GLONET is evaluated by comparing its outputs against GLORYS12 reanalysis products, which serve as the reference. GLORYS12 provides high-quality, observationally informed ocean state estimates, making it an established benchmark for assessing forecast performance in ocean modeling. To contextualize GLONET’s forecasting capabilities within the domain of data-driven modeling, its 10-day forecasts are compared with those of Xihe, another NN-based forecasting system. Both GLONET and Xihe are evaluated using identical initial conditions, the same GLORYS12 reference fields, and over a consistent evaluation period spanning January to July 2024. This setup ensures a fair and uni-

form comparison, enabling an objective assessment of GLONET’s performance relative to other data-driven approaches.

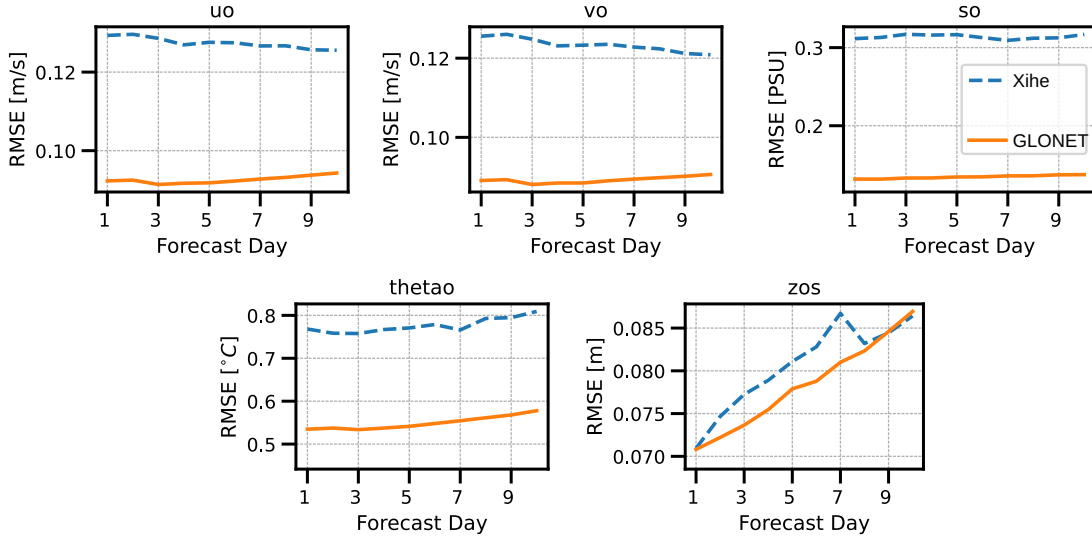


Figure 5: Root Mean Square Error (RMSE) averaged across all vertical levels for 3D variables (U and V currents, temperature, salinity), alongside the RMSE of sea surface height (SSH), for GLONET (orange line) and Xihe (blue line). The calculations cover the period from January to July 2024, using GLORYS12 as the reference dataset. Forecasts are initialized weekly on Wednesdays based on nowcast analyses performed with GLO12, lagging real-time by seven days.

Figure 5 displays the forecasting performance of GLONET and Xihe, expressed in terms of Root Mean Square Error (RMSE). The RMSE is calculated for all forecasted variables over a 10-day forecast window, averaged across all vertical levels for 3D variables, for the period from January to July 2024. In general, GLONET produces highly skillful forecasts as can be seen from the RMSE scores, particularly for ocean currents compared to Xihe, with an improvement of 23%.

A gradual decrease of forecast accuracy over time is observed for GLONET, mainly due to its auto-regressive fashion. Conversely, Xihe exhibits relatively stable performance across all lead times, which can be attributed to its design, where separate models are trained for each specific lead time.

The RMSE scores presented here represent averages across all vertical levels for 3D variables. Since the models operate with distinct vertical level structures, the depth-wise accuracy of forecasts is also investigated. This time, all 10-day forecasted lead times are averaged for the period of January to July 2024, for each vertical level. This allows us to examine the depth-wise accuracy of the forecasts for a more granular understanding of the different models’ performances.

Figure 6 shows the RMSE scores averaged over all lead times for each vertical level, highlighting variations in forecasting accuracy at different depths. This analysis reveals that, in terms of ocean currents, GLONET continues to perform well, with Xihe coming closer to GLONET’s performance in the upper levels, particularly within the mixed layer. For temperature, GLONET performs comparably to Xihe in the intermediate layers. However, GLONET diverges slightly from Xihe at the surface and bottom levels. This depth-dependent analysis provides valuable insights into the models’ forecasting capabilities at different ocean vertical levels, illustrating their strengths and weaknesses across the vertical profile.

In addition to the above analyses, the 10-day forecast accuracy is further evaluated by averaging RMSE across only the common depth levels shared by all models as shown in Figure 7. This analysis reveals that GLONET continues to perform well in predicting ocean currents, closely matching Xihe’s accuracy within the mixed layer. For temperature, GLONET’s performance is similar to Xihe at intermediate depths, while Xihe shows slight advantages at the surface and the bottom layer.

In addition to the analyses presented earlier, a complementary evaluation focusing on the spatial distribution of forecasting errors at the surface level is provided. This analysis aims to assess the regional variability of

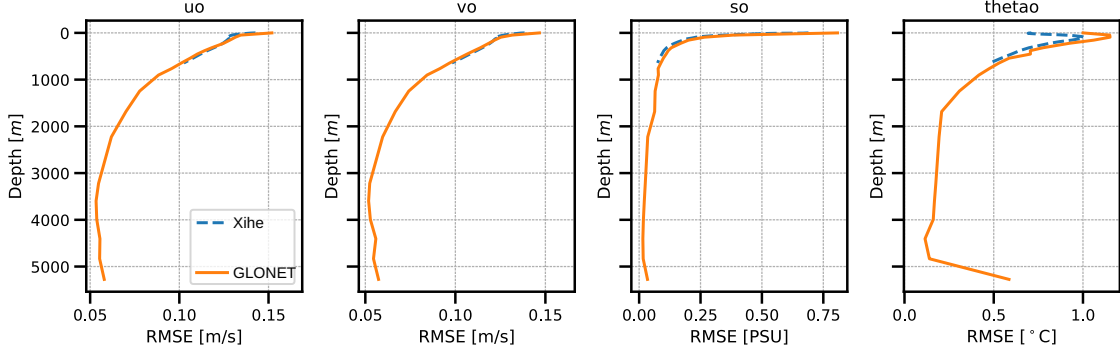


Figure 6: RMSE computed at each depth level and averaged over all lead times for 3D variables (U and V currents, temperature, and salinity) for the GLONET (orange line) and Xihe (blue line) models, covering January to July 2024. GLORYS12 serves as the reference, with 10-day forecasts initialized weekly on Wednesdays from a nowcast analysis performed with GLO12 seven days behind the real-time.

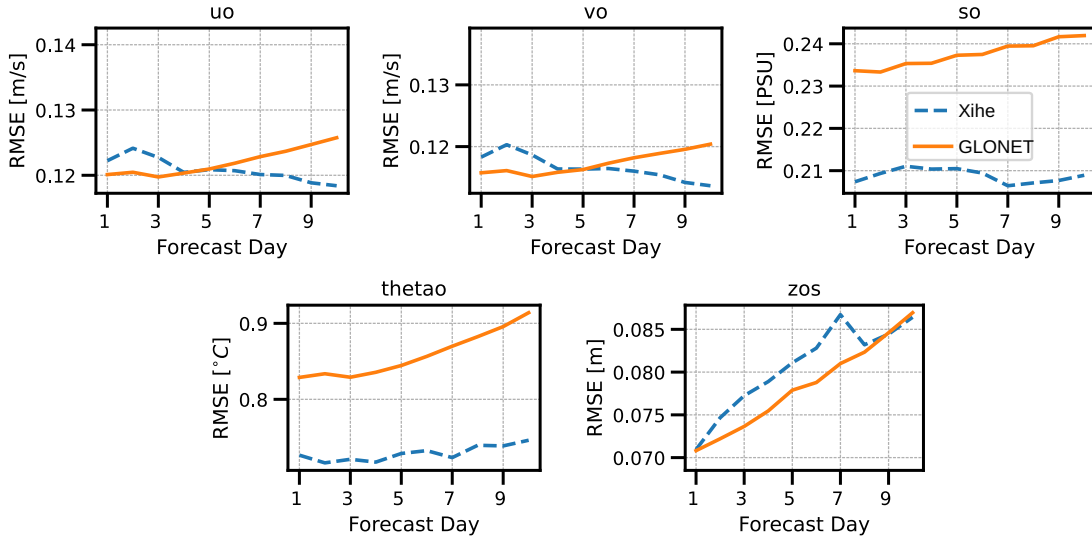


Figure 7: RMSE averaged across common depth levels of 3D variables (U and V currents, temperature, salinity), along with RMSE of sea surface height (SSH) for the GLONET and Xihe models. Calculations span from January to July 2024, using GLORYS12 as the reference. Forecasts are initialized weekly on Wednesdays, following the operational protocol.

model performance, offering insights into how well GLONET and Xihe capture the spatial patterns of ocean dynamics. The spatial RMSE maps at the surface level, shown in Figure 8, reveal that both models, GLONET and Xihe, exhibit a broadly similar regional distribution of errors across all forecasted variables (U and V currents, temperature, and salinity). This consistency in spatial patterns suggests that the primary sources of forecasting error are shared between the models, likely driven by the underlying ocean dynamics and the challenges associated with accurately resolving certain high-variability regions, such as western boundary currents and equatorial zones. While the magnitude of RMSE varies, the spatial consistency underscores the difficulty of forecasting in these regions regardless of the modeling approach.

4.3 Process-oriented evaluations

In this part, GLONET is evaluated using a process-oriented approach, which involves computing derived quantities and analyzing complex processes from the forecasted data. Unlike physics-driven models, which rely on governing equations to represent ocean dynamics, data-driven models are primarily trained to minimize

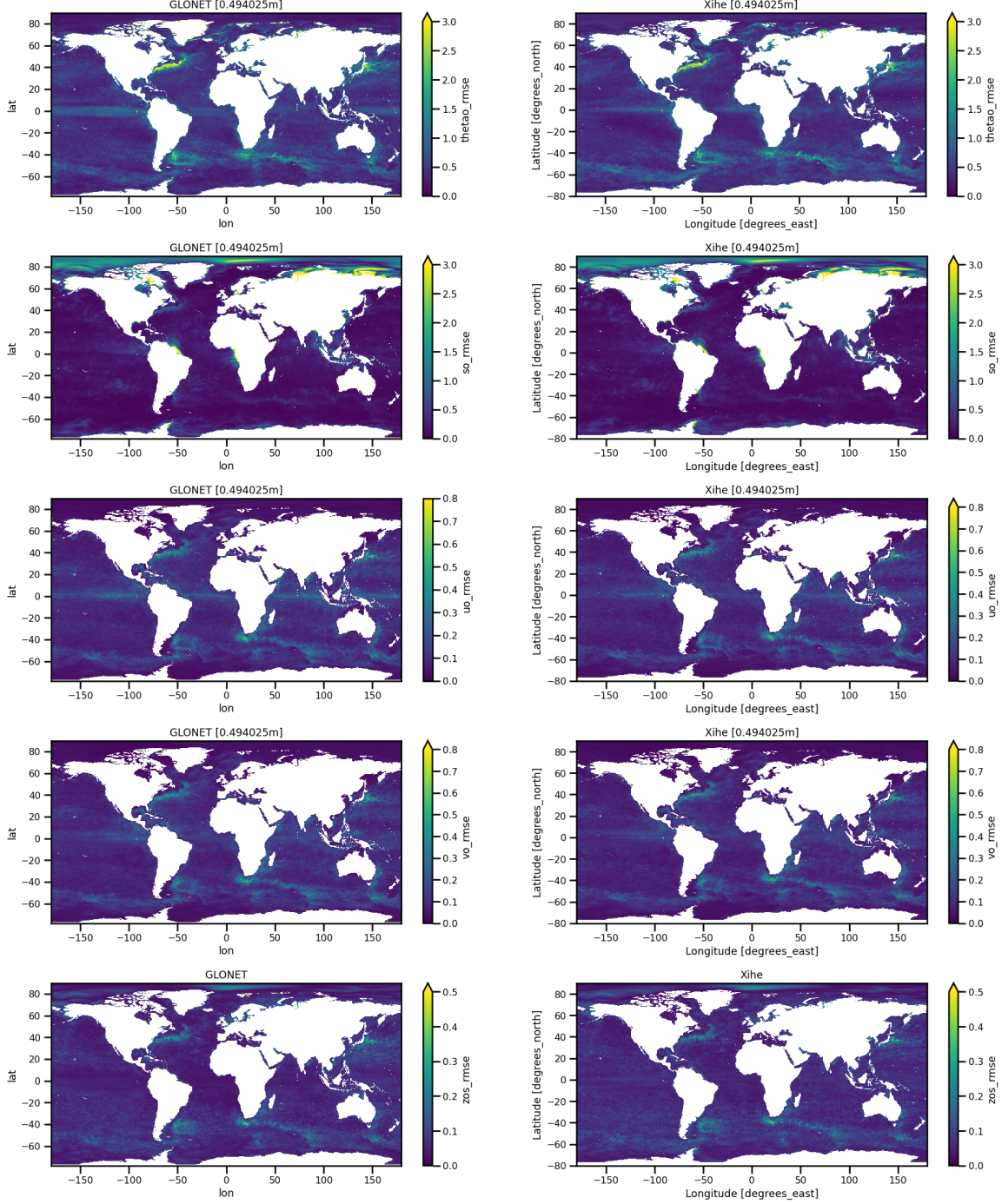


Figure 8: Spatial distribution of RMSE at the surface level for GLONET and Xihe across all forecasted variables (U and V currents, temperature, and salinity and SSH). The RMSE is computed with respect to GLORYS12 over the evaluation period from January to July 2024. These maps provide insight into the regional performance of each model, highlighting areas of stronger or weaker agreement with the reference reanalysis. Results are averaged over all lead times for weekly forecasts initialized on Wednesdays, adhering to the operational protocol.

a loss function, typically based on RMSE. Consequently, neural network-based models are optimized to achieve the best point-wise accuracy but are not inherently guided by the physical laws that govern ocean system. This limitation necessitates additional forms of validation to ensure that the models not only achieve

low error metrics but also respect the underlying dynamics and interactions within the ocean system. By deriving critical oceanographic quantities such as mixed layer depth and geostrophic currents from the models forecasted data, the models abilities to capture realistic physical relationships within the ocean is assessed. Moreover, this process-oriented evaluation provides insight into the internal consistency and coherence across forecasted variables, verifying that the interactions among temperature, salinity, currents, and sea surface height are physically plausible and aligned. Such coherence is crucial for oceanographic forecasting, as the interplay between these variables underpins accurate representation of the complex ocean processes. A lack of alignment could signal inconsistencies that arise from the data-driven models which focus on minimizing RMSE rather than capturing these interdependent dynamics. This dual assessment of derived quantities and variable consistency thus provides a rigorous evaluation of neural-based forecasting systems capacity to not only deliver accurate forecasts on an individual variable basis but also to maintain the integrity of the broader ocean system, offering a holistic view of the models abilities to replicate realistic ocean dynamics.

4.3.1 Derived quantities

In this part two essential physical properties are derived; geostrophic currents [Rhines, 1979] and mixed layer depth (MLD) [de Boyer Montégut et al., 2004] from the forecasted variables to assess the physical consistency, dynamic accuracy, and potential artifacts in the models outputs.

Geostrophic currents are derived from forecasted SSH under the geostrophic approximation:

$$\mathbf{v}(\phi, \theta, t) = gf^{-1}\nabla^\perp\eta(\phi, \lambda, t) \quad (10)$$

where g is the acceleration of gravity, f presents the Coriolis coefficient, and $\eta(\phi, \lambda, t)$ is the sea surface height (SSH), which serves as a noncanonical Hamiltonian for surface velocity. $\perp$ stands for a 90° anticlockwise rotation of the gradient vector, producing a perpendicular flow direction as dictated by geostrophic balance.

Figure 9 presents examples of derived geostrophic currents computed from GLONET, Xihe, and GLORYS12, the latter serving as a reference. GLONET’s derived geostrophic currents show very clean results, reproducing structures and patterns nearly identical to those in GLORYS12. This indicates a high degree of physical consistency in GLONET’s SSH forecasts, capturing the expected gradient flows with minimal distortion. The similarity in structure between GLONET and GLORYS12 further attests to GLONET’s capability in maintaining dynamic accuracy even when evaluated through derived, process-oriented quantities. In contrast, Xihe’s derived geostrophic currents appear notably noisy, with visible artifacts in the output. These artifacts are largely concealed in direct SSH predictions but become evident when calculating the geostrophic current, a process that involves spatial gradients. Since gradient-based calculations can amplify inconsistencies or abrupt changes, Xihe’s noise suggests underlying instabilities or artifacts in the SSH field that would otherwise remain undetected.

Mixed Layer Depth (MLD) is derived from forecasted temperature and salinity profiles and serves as a key indicator of upper-ocean stratification and mixing. MLD is commonly defined based on a density threshold criterion, such that the mixed layer is the depth at which the density difference from the surface equals a specified threshold. The MLD can be approximated as:

$$\text{MLD} = \min \{z \mid \rho_z - \rho_0 \geq \Delta\rho\} \quad (11)$$

where ρ_z represents the density at depth z , ρ_0 is the density at the surface, and $\Delta\rho$ is a threshold value typically set to a small increment (e.g., 0.03 kg/m^3) to capture the mixed layer’s depth relative to surface conditions.

Figure 10 illustrates the derived MLD from GLONET, Xihe, and GLORYS12, with GLORYS12 again serving as a reference. GLONET’s MLD fields demonstrate a high degree of consistency with those of GLORYS12, accurately capturing large-scale spatial patterns of the mixed layer across the global ocean. In most regions, GLONET’s MLD estimates closely match the depth and extent observed in GLORYS12, indicating that GLONET is effectively preserving the dynamic relationship between temperature, salinity, and density. However, in the North Atlantic, where complex mixing processes and deep-water formation occur, GLONET’s MLD representation shows slight discrepancies compared to GLORYS12. This discrepancy suggests that while GLONET captures the broad structure of the mixed layer, there may be challenges in resolving finer-scale vertical mixing processes in regions with intense oceanic convection.

In contrast, Xihe’s derived MLD fields are noticeably noisier, with considerable spatial inconsistencies and a lack of coherent structure, indicating underlying artifacts. These artifacts reflect instabilities in Xihe’s forecasted temperature and salinity fields, leading to inconsistencies in the derived density profiles. Such

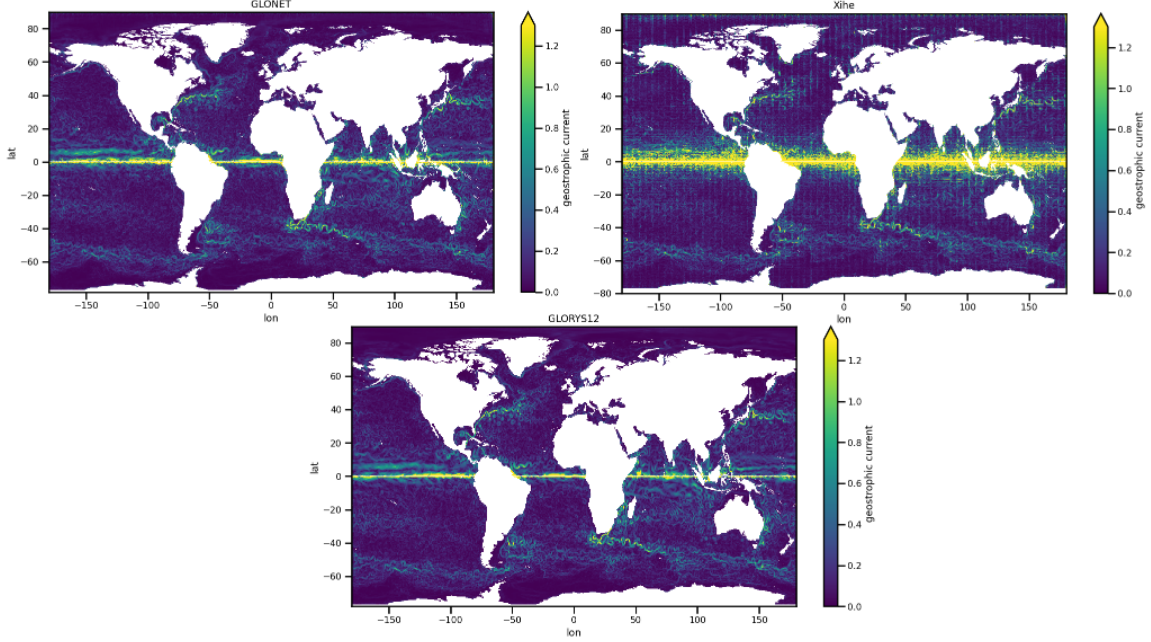


Figure 9: Examples of approximated geostrophic currents using outputs of GLONET and Xihe over the same date (05-01-2024) along with GLORYS12 serving as a reference.

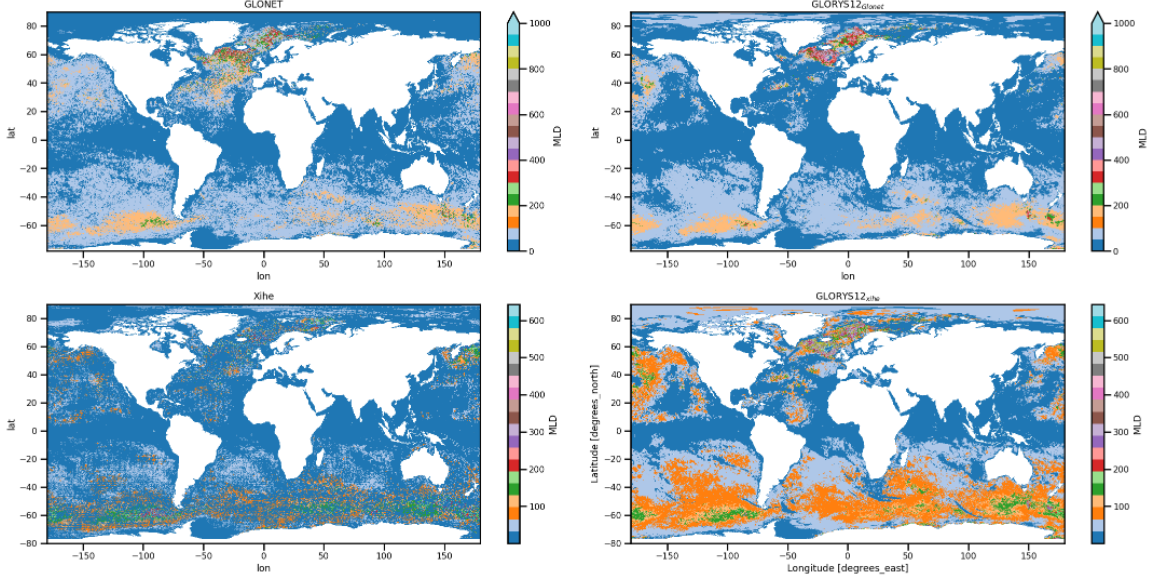


Figure 10: Examples of MLD reconstructed from GLONET and Xihe on 05-01-2024, along with those computed from GLORYS12 with two configurations replicating each time the vertical resolution of GLONET and Xihe.

issues underscore the limitations of models like Xihe in capturing the delicate balance between temperature and salinity required for realistic MLD calculations. The observed noise and irregularity in Xihe’s MLD also point to inconsistencies between the forecasted temperature and salinity fields, which can disrupt the physical consistency of derived variables.

Statistical Analysis of Derived Quantities In addition to visual comparisons, a statistical analysis is conducted to further quantify the accuracy of GLONET and Xihe in predicting derived quantities over a 10-days forecast horizon, averaged across the entire evaluation period from January to July 2024. Specifically,

the geostrophic currents and mixed layer depth (MLD) are computed from GLONET’s and Xihe’s forecast outputs, comparing each to the GLORYS12 reanalysis, which serves as the reference standard. This approach mirrors the RMSE evaluations applied to primary variables, facilitating a consistent comparison of model performance across derived physical properties.

The results (Figure 11) show that GLONET maintains low RMSE values for both geostrophic currents and MLD throughout the forecast period, achieving close alignment with GLORYS12. This accuracy underscores GLONET’s capacity to produce forecasts that are not only quantitatively accurate but also dynamically consistent with observed ocean state dynamics.

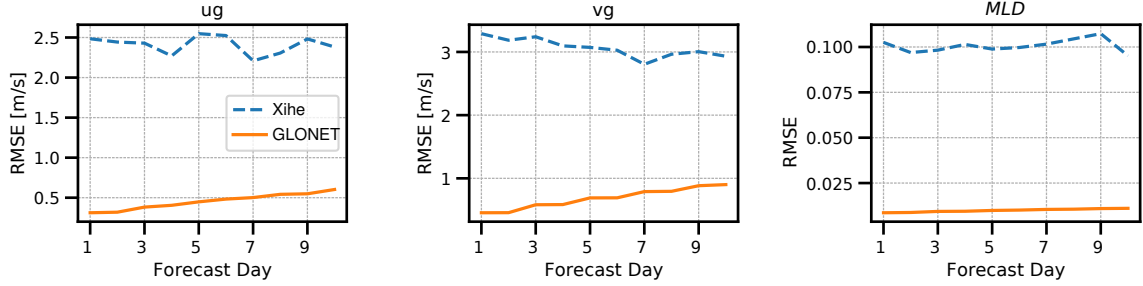


Figure 11: Root Mean Square Error (RMSE) for the reconstructed geostrophic current and MLD for both GLONET, Xihe models. Calculations span from January to July 2024, using GLORYS12 as the reference. Geostrophic current and MLD are computed based on forecasts that are initialized weekly on Wednesdays, following the operational protocol.

In contrast, Xihe displays significantly higher RMSE values for both geostrophic currents and MLD, indicating substantial deviations from GLORYS12. For geostrophic currents, the elevated RMSE suggests notable discrepancies in the SSH-derived fields, potentially due to unresolved artifacts or inconsistencies. Similarly, the high RMSE in MLD forecasts suggests challenges in preserving coherence between temperature and salinity, pointing to physical limitations in Xihe’s derived fields.

This statistical comparison highlights GLONET’s ability to deliver dynamically consistent forecasts across derived variables, with low RMSE values that reflect its skill in reproducing complex ocean dynamics. On the other hand it also reveals a limitation in using RMSE as the primary metric for evaluating data-driven models, as RMSE is not sensitive to small-scale inconsistencies or artifacts that affect the derived fields. RMSE measures point-wise accuracy and fails to capture the spatial continuity and coherence critical for processes like geostrophic flow derivation. This highlights the importance of process-oriented validation metrics. By deriving geostrophic currents from SSH, a more comprehensive evaluation of each model’s performance in preserving the integrity of physical fields is gained. While deriving the MLD offers insights into each model’s ability to preserve physically consistent interactions between forecasted variables, particularly in capturing coherent and artifact-free stratification in the upper ocean.

In addition to the analyses presented earlier, a complementary evaluation focusing on the spatial distribution of forecasting errors for derived quantities is provided, specifically geostrophic currents and mixed layer depth (MLD). This analysis aims to assess the regional variability of model performance, offering insights into how well GLONET and Xihe capture the spatial patterns of these derived physical properties. The spatial RMSE maps, shown in Figure 12, reveal a stark contrast between the models. GLONET demonstrates a regional error distribution with coherent spatial patterns and relatively low errors across most regions. In contrast, Xihe shows significantly noisier error distributions, with high RMSE values scattered across all regions. This noisiness reflects inconsistencies in Xihe’s forecasts, potentially linked to artifacts in the data and weaker coherence between variables such as temperature, salinity, and SSH. These results underscore GLONET’s ability to preserve physical consistency in derived quantities.

4.4 Diagnostic Variables

To complement the analysis of derived quantities, diagnostic variables that provide deeper insights into the intrinsic dynamics and energy-cascading properties of the forecasting models are explored. These diagnostics serve as critical tools for evaluating how well the models replicate fundamental ocean processes, including rotational dynamics and mesoscale variability. In this context, the output of GLO12 is incorporated, not as a

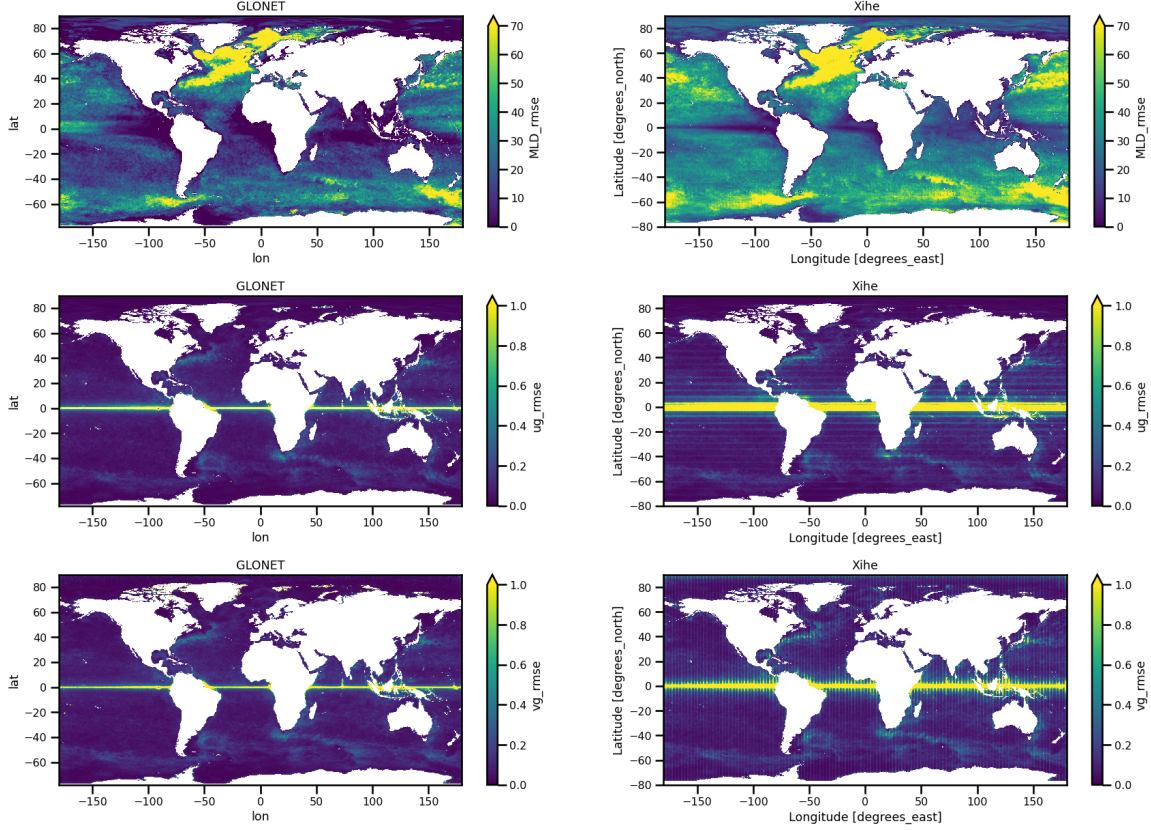


Figure 12: Spatial distribution of RMSE at the surface level for derived geostrophic currents and mixed layer depth (MLD) from GLONET and Xihe. The RMSE is computed with respect to GLORYS12 over the evaluation period from January to July 2024. These maps provide insight into the regional performance of each model in capturing derived physical properties, highlighting areas of stronger or weaker agreement with the reference reanalysis. Results are averaged over all lead times for weekly forecasts initialized on Wednesdays, adhering to the operational protocol.

benchmark, but as a reference framework to evaluate the fidelity of neural network-based models like GLONET and Xihe in reproducing key oceanic features. Particular emphasis is placed on vorticity and mesoscale eddies are investigated, both of which are directly tied to the dynamic integrity and physical realism of the modeled ocean states.

Vorticity Analysis In ocean dynamics, vorticity is a key diagnostic variable that characterizes the rotational motion of fluid parcels, offering insights into energy distribution, flow stability, and turbulence. Unlike derived quantities such as mixed layer depth (MLD) or geostrophic currents, which assess inter-variable coherence, vorticity delves into the system’s internal dynamics and cascade-preserving capabilities. The vorticity, denoted as ω , is calculated from the zonal (u) and meridional (v) components of the currents as follows:

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}, \quad (12)$$

where $\frac{\partial v}{\partial x}$ represents the spatial derivative of the meridional velocity with respect to longitude, and $\frac{\partial u}{\partial y}$ is the spatial derivative of the zonal velocity with respect to latitude. This formulation captures the rotational characteristics of ocean currents, serving as a diagnostic for the conservation of small- and large-scale structures within the forecast.

As shown in Figure 13, the vorticity fields for GLONET, GLO12, and Xihe over the Gulf Stream region reveal critical differences in dynamic fidelity. GLONET and GLO12 maintain robust coherence in vorticity across both small and large-scale structures, reflecting effective cascade-preserving capacities over the entire 10-day

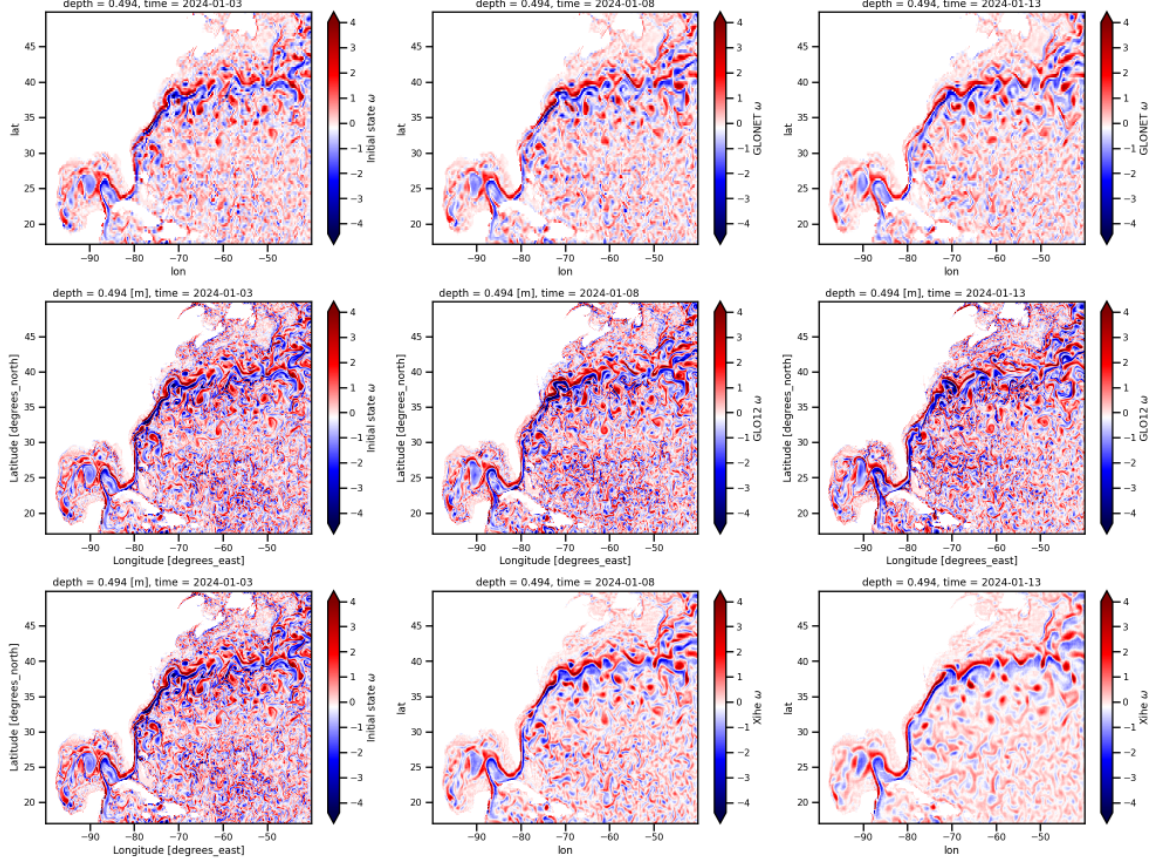


Figure 13: Evolution of vorticity over a 10-day forecast period for GLONET, GLO12, and Xihe, displayed over the Gulf Stream region. GLONET and GLO12 demonstrate stable energy-cascading and retain coherent rotational structures, whereas Xihe exhibits energy cascade attenuation, leading to progressive loss of small-scale features over time.

forecast period. This ability to preserve the energy transfer across scales is crucial for accurately capturing mesoscale phenomena and the energetic dynamics of ocean currents.

In contrast, Xihe displays a significant energy cascade attenuation as the forecast progresses, with the fields appearing overly smooth by day 10. While large-scale rotational features persist, the loss of small-scale energy indicates an underlying instability or dissipation issue. This pattern resembles decaying turbulence, where energy cascades out of the system, undermining the representation of critical mesoscale dynamics. This behavior can be attributed to Xihe’s architectural design, which employs independently trained models for each lead time. While this design stabilizes point-wise RMSE metrics, it neglects dynamic continuity, leading to the observed decay in fine-scale vorticity structures. This strategy might also neglects continuity across vertical levels as demonstrated in Figure 18 in Appendix.

Energy Cascading Analysis: Beyond the visual analysis of cascade attenuation, where a progressive loss of small-scale features is observed over time, the energy dynamics is further investigated using a quantitative approach. To assess the retention of small-scale dynamics over time, the energy distribution, derived from the vorticity field, is analyzed across spatial scales. This way the models’ ability to preserve fine-grained dynamics critical for accurate ocean forecasting is evaluated.

The vorticity fields is analyzed in the spectral domain to determine their energy content at different spatial scales. Specifically, the vorticity field $\omega(x, y)$ is subjected to a two-dimensional discrete Fourier transform to obtain its spectral representation $\hat{\omega}(k_x, k_y)$, where k_x and k_y represent the zonal and meridional wavenumbers, respectively. Then the power spectrum is computed as $|\hat{\omega}(k_x, k_y)|^2$, quantifying the energy associated with each spectral component.

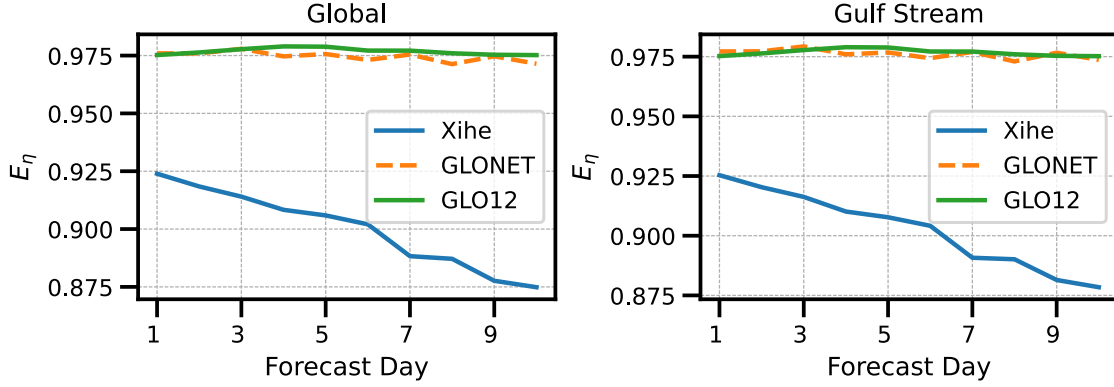


Figure 14: Comparison of small-scale energy fraction over 10-days forecast period for Xihe, GLONET, and GLO12 models. The analysis was conducted from vorticity fields over the Gulf Stream (right) and the entire globe (left) over the evaluation period from January to July 2024.

To analyze the energy distribution in a more interpretable form, the two-dimensional power spectrum is radially averaged in wavenumber space, resulting in the one-dimensional energy spectrum $E(k)$, where $k = \sqrt{k_x^2 + k_y^2}$. This spectrum represents the energy density as a function of total wavenumber, enabling the characterization of energy across spatial scales.

Small-scale energy content is defined as the sum of $E(k)$ over all wavenumbers $k > k_c$, where the cutoff wavenumber k_c corresponds to a physical length scale L_c (e.g., $L_c = 100$ km), and the total energy is computed by summing $E(k)$ across all wavenumbers. The fraction of energy associated with small scales is then defined as:

$$E_\eta = \frac{\sum_{k > k_c} E(k)}{\sum_k E(k)}$$

This dimensionless fraction quantifies the proportion of total energy contained in small spatial scales relative to the entire spectrum.

By examining the small-scale energy fraction over the 10-days forecast horizon, the ability of each model to maintain the energy cascade characteristic of dynamic fidelity is evaluated. This analysis is conducted for 10-days forecasts spanning the period from January to July 2024, covering both global domains and the Gulf Stream region. The results consistently revealed that both GLONET and GLO12 maintained stable small-scale energy fractions throughout the forecast period, demonstrating their capacity to preserve fine-scale features and sustain energy consistency. Notably, the stability of these fractions was observed across both the global domain and the Gulf Stream region, underscoring the robustness of GLONET and GLO12 in preserving small-scale dynamics.

Conversely, Xihe exhibited a marked decline in the small-scale energy fraction over the forecast horizon, indicative of its inability to sustain the energy cascade. This pattern, consistent across both the global domain and the Gulf Stream region, aligns with the observed decay of vorticity structures in Xihe forecasts. These findings highlight Xihe's broader challenges in maintaining dynamical coherence and preserving the intricate coupling between scales, essential for accurate ocean forecasting.

Mesoscale Eddies Mesoscale eddies, integral to ocean dynamics, were analyzed to assess the models' capacity to represent coherent structures. Eddies were identified using an Eulerian approach, based on detecting closed SSH contours with a monotonic increase or decrease towards a central extremum. To ensure consistency across models, no smoothing was applied to the SSH fields, and a convexity criterion of at least 0.9 was imposed to confirm eddies coherence. Over the 10-day forecast horizon, no significant temporal evolution in eddy distribution is expected, therefore this analysis did not involve tracking individual eddies. Instead, it focuses on detecting spatial patterns at each lead time.

The analysis shows that GLONET and GLO12 perform comparably well, accurately capturing the spatial distribution of mesoscale eddies over the 10-day forecast period. Figure 15 illustrates the regional distribution of detected eddies, revealing that both GLONET and GLO12 align closely with the expected patterns of mesoscale variability. This consistency underscores their ability to maintain the SSH gradients necessary for eddy detection and to preserve dynamic integrity across lead times.

Conversely, Xihe exhibits significant shortcomings, frequently failing to detect eddies or misrepresenting their spatial structure. This deficiency highlights Xihe’s difficulty in maintaining coherent SSH gradients, which is essential for identifying mesoscale features. These results emphasize the critical role of dynamic consistency in accurately representing mesoscale processes. Both vorticity and mesoscale eddies serve as diagnostic variables that reveal how well forecasting models replicate the fine-scale and energetic dynamics of ocean circulation.

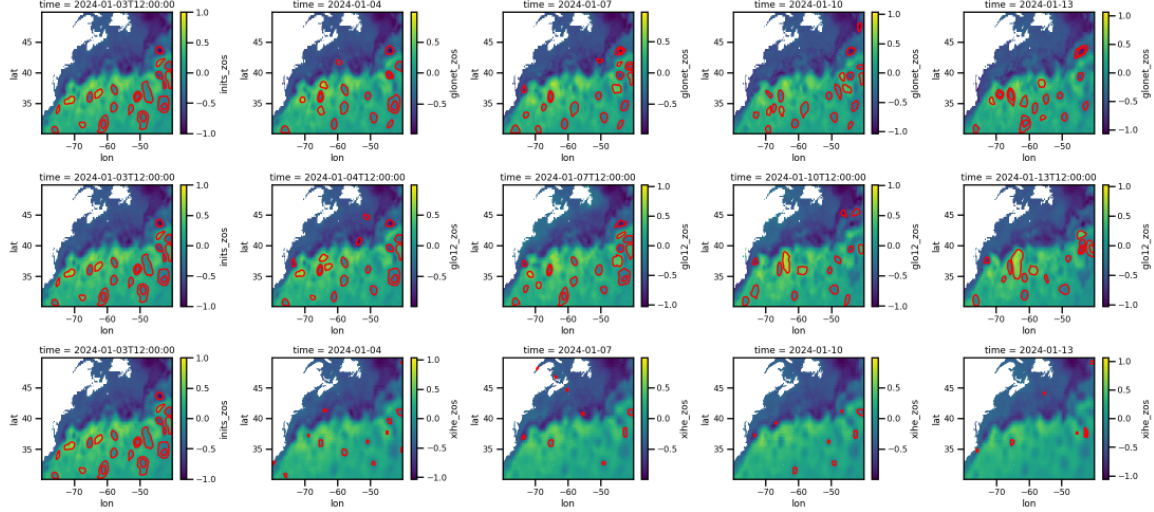


Figure 15: Evolution of eddies over a 10-day forecast period for GLONET, GLO12, and Xihe, displayed over the Gulf Stream region. First column represent the eddies extracted from the initial state inputted to all the models.

4.4.1 Lagrangian analysis

In this subsection, the process-oriented accuracy of GLONET is assessed by analyzing Lagrangian trajectories over a forecast horizon of 10 days, using daily forecast outputs from January to July 2024. This approach offers insight into the model’s ability to capture the advection of ocean particles accurately over time, which is critical for applications that rely on realistic flow and transport simulations.

Let’s consider the ocean currents field:

$$\mathbf{v}(\mathbf{x}, t), \quad \mathbf{x} \in \mathbb{R}^2, \quad t \in [t_0, t_f] \quad (13)$$

and its associated ordinary differential equation:

$$\dot{\mathbf{x}} = \mathbf{v}(\mathbf{x}, t), \quad \mathbf{x} \in \mathbb{R}^2, \quad t \in [t_0, t_f] \quad (14)$$

where $\mathbf{v}$ the U and V components of ocean currents, defined on a possibly time-dependent spatial domain $\mathcal{U}(t) \in \mathbb{R}^2 \times [t_0, t_f]$.

Lagrangian trajectories are defined as:

$$\mathbf{x}(t_f, t_0, \mathbf{x}_0) = \mathbf{x}_0 + \int_{t_0}^{t_f} \mathbf{v}(\mathbf{x}(\tau), \tau) d\tau \quad (15)$$

Following the above, Lagrangian trajectories of the different models are computed by setting up $\mathbf{v}$ each time to ocean current forecasts from GLONET, GLO12 and Xihe, with the flow integration being carried out

following a fourth-order Runge-Kutta method with variable step-size. For each model, particles initialized at identical locations across the red boxes of Figure 16 are tracked, and Euclidean distances between the modeled trajectories and those from GLORYS12 are calculated by setting up t_0 to the first lead-day forecast and t_f to the 10th lead-day. This Euclidean distance metric provides a quantitative assessment of each model’s accuracy in reproducing the physical transport patterns present in the reanalysis data.

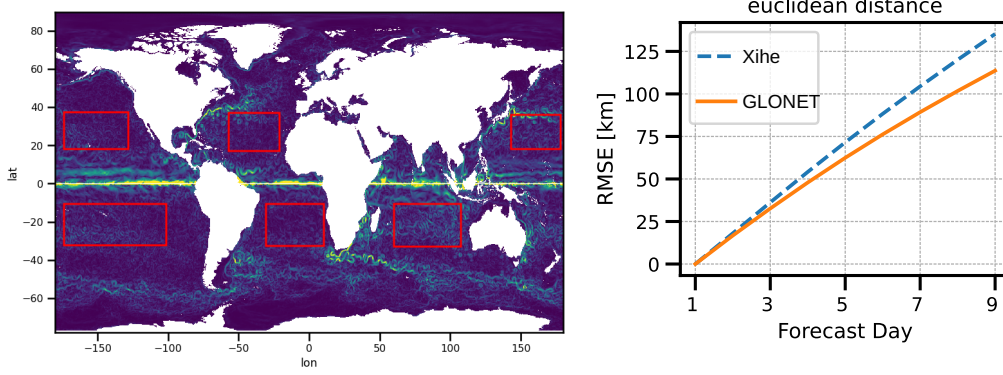


Figure 16: (Left) Red boxes highlight the areas where particles were initiated. (Right) Divergence of GLONET and Xihe’s Lagrangian trajectories from those of GLORYS12. Trajectories reconstruction is done based on the the weekly forecasts span from from January to July 2024.

As shown in Figure 16, GLONET consistently outperforms Xihe, maintaining closer proximity to the GLORYS12-derived trajectories. At the 10-day lead time, GLONET reduces the Euclidean distance error by approximately 25 km. This improvement reflects GLONET’s enhanced capacity to capture accurate flow dynamics and oceanic transport structures over extended forecast horizons.

Comparison of Geostrophic and Full Lagrangian Trajectories To further evaluate the dynamical coherence within each model, Lagrangian trajectories are reconstructed using both geostrophic currents (derived from SSH) and full currents for GLONET, GLO12, and Xihe. By comparing the Euclidean distances between these geostrophic and full trajectories, the degree of alignment between SSH and current fields are assessed, offering insight into each model’s capability to represent cohesive, physically consistent flow dynamics.

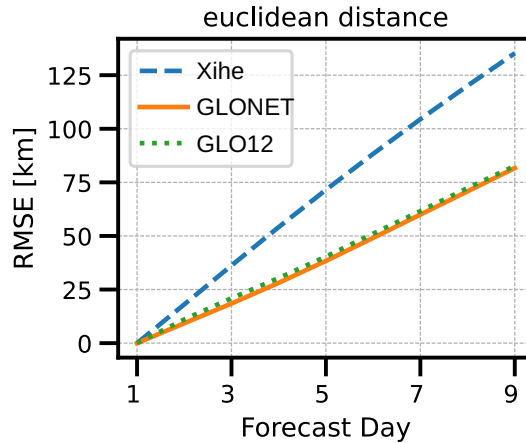


Figure 17: Divergence of geostrophic Lagrangian trajectories from those computed using full current each time for GLO12, GLONET and Xihe. Trajectories calculations is done based on the weekly forecasts of ocean currents span from January to July 2024 along with geostrophic currents approximated from the forecasted SSH.

The results, illustrated in Figure 17, show that GLONET and GLO12 exhibit consistent RMSE evolution over the 10-day forecast period, with a stable and generally smaller Euclidean distance between geostrophic and full trajectories. This close alignment indicates that both models maintain a high degree of coherence

between SSH and current predictions, reflecting an ability to generate forecasts with internally consistent surface and total flow fields. Conversely, Xihe demonstrates a rapid divergence between geostrophic and full current-based trajectories, evident in the increasing Euclidean distance over time. This discrepancy suggests a lack of coherence between Xihe’s SSH and current fields, likely due to artifacts that surface in derived geostrophic currents. The divergence highlights Xihe’s limitations in producing dynamically consistent outputs across interconnected variables.

5 Discussion and Summary

A comprehensive evaluation of GLONET, a novel data-driven ocean forecasting model, is conducted by comparing its performance against two systems: GLO12, a physics-driven operational model, and Xihe, an AI-based forecasting approach. The assessment encompassed multiple dimensions, including traditional point-wise error metrics, process-oriented analyses, diagnostic variables, derived physical quantities, and observation-based evaluations using the IV-TT Class 4 Framework. This multidimensional perspective highlights GLONET’s strengths and limitations relative to established approaches.

The CLASS4 framework provided an independent evaluation of the systems’ performance by comparing model outputs directly against in situ and satellite observations. Analyses of temperature and salinity in the 5–100 m layer, SST from drifting buoys, and SLA from satellite altimeters revealed that GLONET consistently outperformed Xihe across most metrics and forecast lead times. In particular, GLONET demonstrated superior accuracy in SLA forecasts, aligning closely with GLO12 for intermediate lead times (days 5–9). However, GLO12 displayed an advantage in SST forecasting, especially in tropical regions, underscoring the challenges of accurately resolving surface variables in data-driven models. These CLASS4-based evaluations emphasized GLONET’s capacity to produce forecasts comparable to operational standards while highlighting areas for further refinement.

Point-wise evaluations, conducted using GLORYS12 as the reference dataset, assessed the models’ ability to generate forecasts consistent with the data used during training. GLONET exhibited strong performance in predicting ocean currents and comparable accuracy in temperature and salinity forecasts, reflecting its capability to capture large-scale ocean dynamics. GLONET also experienced a gradual decline in forecast accuracy with increasing lead times, characteristic of autoregressive models, whereas Xihe maintained stable error levels due to its lead-time-specific training strategy. This stability, however, came at the expense of inter-variable consistency and dynamic coherence.

Extending beyond point-wise assessments, process-oriented analyses revealed critical aspects of model performance that are often overlooked in traditional metrics. For instance, GLONET demonstrated a strong ability to derive geostrophic currents from forecasted SSH, maintaining physical consistency and dynamic coherence akin to GLORYS12. Xihe, in contrast, exhibited significant noise, exposing artifacts in its derived quantities, stemming from its RMSE-focused training strategy. Similarly, GLONET’s mixed layer depth (MLD) estimates closely matched GLORYS12, except for localized discrepancies in regions such as the North Atlantic, where finer-scale adjustments to temperature and salinity stratification might be required. These results highlight GLONET’s ability to maintain consistency between temperature and salinity, a critical requirement for accurately representing ocean stratification. Xihe’s MLD predictions, on the other hand, were plagued by noise and inconsistencies, further emphasizing the need for enhanced regularization in AI-driven approaches.

Diagnostic analyses further evaluated the models’ fidelity in preserving dynamic consistency and fine-scale structures. Vorticity provided a direct measure of the rotational motion within the models, serving as a diagnostic for energy cascading and dynamic coherence. Both GLONET and GLO12 successfully preserved vorticity structures over the Gulf Stream region, maintaining coherent small and large-scale features over the 10-day forecast period. In contrast, Xihe displayed a progressive loss of small-scale features, transitioning toward a state resembling decaying turbulence by day 10. This behavior stems from Xihe’s architectural design, which uses independently trained models for each lead time, neglecting dynamic continuity and leading to the observed degradation.

Lagrangian trajectory analyses provided additional insights into dynamic accuracy. Particle tracking over a 10-day forecast horizon highlighted GLONET’s superior ability to maintain coherence across time and space, outperforming Xihe by approximately 25 km in trajectory accuracy. Both GLONET and GLO12 exhibited consistent alignment between geostrophic and full currents, reflecting a strong coupling between SSH and surface velocity fields. Xihe, however, displayed rapid divergence between these trajectory types, revealing weaknesses in its physical coupling.

Mesoscale eddies, critical to ocean energy transport and mixing, were identified using closed SSH contours with a convexity criterion of 0.9, ensuring fair comparison across models without smoothing the SSH fields. GLONET and GLO12 demonstrated strong consistency in eddy detection, capturing the spatial distribution of mesoscale features accurately over the forecast horizon. In contrast, Xihe often failed to detect eddies or misrepresented their structure, reflecting deficiencies in maintaining the SSH gradients necessary for identifying coherent eddy features. These results emphasize the importance of dynamic consistency and structural integrity for accurately resolving mesoscale phenomena.

To complement these analyses, spatial RMSE maps of surface variables and derived quantities provided a regional perspective on model performance. GLONET and GLO12 exhibited nearly identical spatial error distributions, suggesting that both models are influenced by similar physical challenges, such as resolving western boundary currents and equatorial dynamics. Xihe, however, showed a more erratic spatial error pattern, characterized by pervasive noise across derived quantities. This disparity underscores the value of incorporating physical constraints into data-driven models to ensure spatial coherence and reduce artifacts in regions with high variability.

In summary, this study underscores the potential of GLONET as a next-generation ocean forecasting model that effectively bridges the gap between traditional physics-based systems and purely data-driven approaches. By capturing key ocean dynamics, ensuring inter-variable consistency, and producing forecasts competitive with operational standards, GLONET demonstrates its capacity to serve as a robust forecasting tool. Nevertheless, localized discrepancies and the inherent limitations of purely data-driven methodologies highlight the need for further refinements. This work also illustrates the necessity of comprehensive evaluation frameworks in ocean forecasting. Traditional metrics like RMSE, while informative, are insufficient to fully capture the dynamical fidelity and inter-variable coherence required for reliable forecasts. Process-oriented assessments, diagnostic variables and derived quantity analyses provide crucial insights into data-driven models performance, ensuring that forecasts not only minimize errors but also respect the fundamental dynamics of the ocean system. Together, these findings pave the way for advancing data-driven ocean forecasting and offer a foundation for future improvements in methodology and application.

Data availability statement

GLONET [1/4°] experimental daily forecast are accessible through the European Digital Twin Ocean (DTO) platform EditO <https://www.edito.eu/>.

GLORYS12 and GLO12 products are available in Copernicus Marine Data Store <https://data.marine.copernicus.eu/>

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Author Contribution Statement

A.E conceived the study, developed the model, conducted the simulations/experiments, and wrote the original draft of the manuscript. Q.G built the pre-operational pipeline for GLONET's integration. C.R conducted the Class-4 evaluation. All authors discussed the results and contributed to the final manuscript.

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Appendix

Vertical sections evaluation

The vertical sections of temperature, salinity, and ocean currents, depicted in Figure 18, provide a detailed comparative assessment of the models in the Gulf Stream region at a latitude of $40.25^{\circ}N$. Both GLONET and GLO12 demonstrate smooth and continuous vertical distributions, highlighting their ability to accurately capture vertical coherence and dynamic consistency across the water column. These results reflect their robust handling of vertical gradients and fine-scale patterns essential for representing ocean stratification and circulation dynamics. In contrast, Xihe exhibits a distinct discontinuity around the 100-meter depth, particularly evident in the ocean current profiles. This artifact arises from its architectural design, which employs two independent models: one optimized for depths between the surface and 100 meters, and another for depths from 100 to 600 meters. While temperature and salinity fields are less impacted, showing minor inconsistencies at the model transition depth, the pronounced discontinuity in ocean currents highlights the challenges associated with maintaining smooth transitions across depth ranges in data-driven frameworks that rely on segmented modeling.

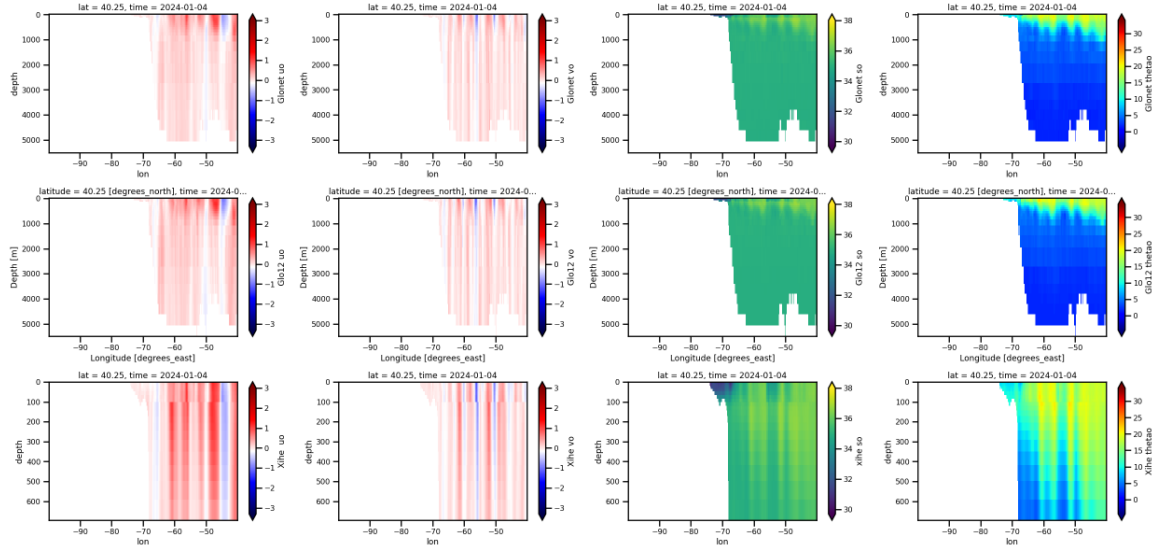


Figure 18: Vertical sections of the Gulf Stream for 3D variables (U and V currents, temperature, and salinity) from GLO12, GLONET, and Xihe on the same date. Notably, Xihe shows a pronounced 100 m depth shift, particularly evident in the zonal (U) and meridional (V) current components.